# Nayoung Ku

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## Education

**Handong Global University** Feb. 2023 – Feb. 2026
*B.Sc. in Life Science and Applied Artificial Intelligence (cum laude, GPA: 4.0/4.5)* *Pohang, South Korea*

- **Bachelor's Thesis:** Language-Model-Based RNA-seq Data ETL Workflow for Ovarian Cancer Classification Model

**National Institute for Lifelong Education** Feb. 2021 – Feb. 2023
*B.L.A. in Psychology, Bachelor's Degree Examination for Self-Education (GPA: 4.0/4.3)* *Seoul, South Korea*

## Honors & Awards

**3rd Place, Daewoong Pharmaceutical President's Award (out of 990 teams)** Oct. 2024
*JUMP AI Drug Discovery Challenge – Organized by KPBMA & Sponsored by the Ministry of Health and Welfare*

**Big Data Seagull Scholarship** Spring 2024, Fall 2024
*Merit-based scholarship, certified by HGU School of Applied Artificial Intelligence*

## Research Experience

**Undergraduate Researcher → Researcher** Aug. 2024 – Present
*Handong Global University – Advisor: Prof. Jaeyoung Chun* *Pohang, South Korea*

- Exploring gene regulatory network-informed approaches to predict **transcriptomic shifts** in response to **genetic perturbation (CRISPRi)**, aiming for zero-shot prediction on unseen cell lines
- Standardizing heterogeneous **Perturb-seq** datasets and building reproducible model training/evaluation pipelines
- Developed a diffusion-based probabilistic demultiplexing pipeline for hashed scRNA-seq data that learns the underlying signal distribution without hard-threshold cutoffs
- Constructed a high-quality annotated training dataset for **circulating tumor cell (CTC)** detection in unstained blood **microscopy images** and conducted a systematic literature review on CTC morphology
- Initiated and led an **IC50 prediction project** for IRAK4 inhibitors from chemical structure (SMILES), generated molecular descriptors, and designed feature selection methods for ML regression models
- Achieved **top 3 finalist** status at a national AI drug discovery challenge (see Honors & Awards) and **presented the poster at AI Pharma Korea Conference 2024**

**Visiting Undergraduate Researcher** Jun. 2025 – Jul. 2025
*University of Nevada, Las Vegas – Advisor: Prof. Mingon Kang (DataX Lab)* *Nevada, United States*

- Developed a lightweight **patch tiling and filtering algorithm** for **whole slide images (WSIs)** from the **TCIA CMB-LCA**, reducing raw data volume by **80%** (128GB → 25GB), significantly lowering computational overhead
- Built a hybrid deep learning framework integrating **Vision Transformer** embeddings with an **LSTM**-based classifier for large-scale, imbalanced tumor classification across WSIs

**Undergraduate Researcher** Jan. 2025 – Apr. 2025
*Seoul National University – Advisor: Prof. Howard Lee (CCADD)* *Seoul, South Korea*

- Contributed to an **NLP research project commissioned by Korea's Ministry of Food and Drug Safety (MFDS)**, building a **Korean–English parallel sentence corpus** of pharmaceutical regulatory documents
- Standardized data collection, annotation, and management protocols across a multi-source document corpus, establishing a reusable curation pipeline for the research team, and designed a sentence alignment algorithm for automated parallel corpus construction from heterogeneous document sources

**Undergraduate Research Intern** Mar. 2024 – Nov. 2024
*Handong Global University – Advisor: Prof. Taejin Ahn (BioData Lab)* *Pohang, South Korea*

- Analyzed clinical **Tumor-Educated Platelet (TEP) RNA-seq** datasets to identify **ovarian and colorectal cancer biomarkers**, applying feature selection and hierarchical clustering to isolate signal from high data sparsity
- Developed a **transformer-based preprocessing framework** encoding raw RNA sequences of TEPs directly into embeddings without traditional count-matrix construction and achieved **AUC 0.71** in ovarian cancer classification

## Presentations

**N. Ku**, J. Kim. "No Cell Left Behind: Recovering Discarded Cells via Self-Supervised Diffusion Denoising." *Bio-AI Capstone Design Fair*, Handong Global University, Pohang, South Korea, Dec. 2025.

**N. Ku**. "Diagnosing Cancer with Genomic Context." *Capstone Design in Life Science Fair*, Handong Global University, Pohang, South Korea, Nov. 2024.

**N. Ku**, S. Kwon, Y. Choi. "IRAK4 IC50 Inhibition Prediction." *AI Pharma Korea Conference 2024*, Seoul, South Korea, Oct. 2024. (Poster; invited as a winning team of the JUMP AI Drug Discovery Challenge)

## Teaching Experience

**Introduction to AI and Machine Learning, *Teaching Assistant*** Aug. 2025
*Addis Ababa Science and Technology University – Instructor: Prof. Dai Yon Cho* *Addis Ababa, Ethiopia*

**Big Data Modeling and Platforms, *Teaching Assistant*** Mar. 2025 – Jun. 2025
*Handong Global University – Instructor: Prof. Jaeyoung Chun* *Pohang, South Korea*

**AI Application Programming, *Teaching Assistant*** Mar. 2025 – Jun. 2025
*Handong Global University – Instructor: Prof. Sangsan Lee* *Pohang, South Korea*

## Technical Skills

**Languages**: Python, R, SQL, Bash
**ML/DL**: PyTorch, scikit-learn, Weights & Biases, MLflow, Optuna, HuggingFace
**Bioinformatics**: Scanpy, scRNA-seq Analysis, Bulk RNA-seq, Histopathology & Microscopy Image Analysis
**Infrastructure**: Git, GitHub, Slurm, LSF, MIG-partitioned GPU clusters, Linux

## Community Involvement

**R Programming Camp, *Teaching Assistant*** Jan. 2026
*Mentored high school students in data analysis and visualization using R* *Pohang, South Korea*

**Handong Office of International Affairs, *Exchange Student Mentor*** Aug. – Dec. 2024
*Assisted exchange students with adaptation to Korean culture and residential college* *Pohang, South Korea*

**Incheon Beauty and Art High School Python Camp, *Teaching Assistant*** Aug. 2023
*Supported Python programming instruction for elementary to high school students* *Incheon, South Korea*